

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 50
- max_units: 150
- units_step: 50
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh']
- optimizers: ['adam']

Tuning:

- max_epochs: 50
- hyperband_iterations: 3

SimpleRNN Multi-Step Training Report

Report Generated: 2025-05-04 11:43:54

Training Start: 2025-04-29_17-16-08

Training End: 2025-04-29_17-23-46

Training Duration: 7.64 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 10.3405

MSE: 112.9776

RMSE: 10.6291

R2 Score: -17.6802

Multi-Step Forecast:

Forecast Steps: 20

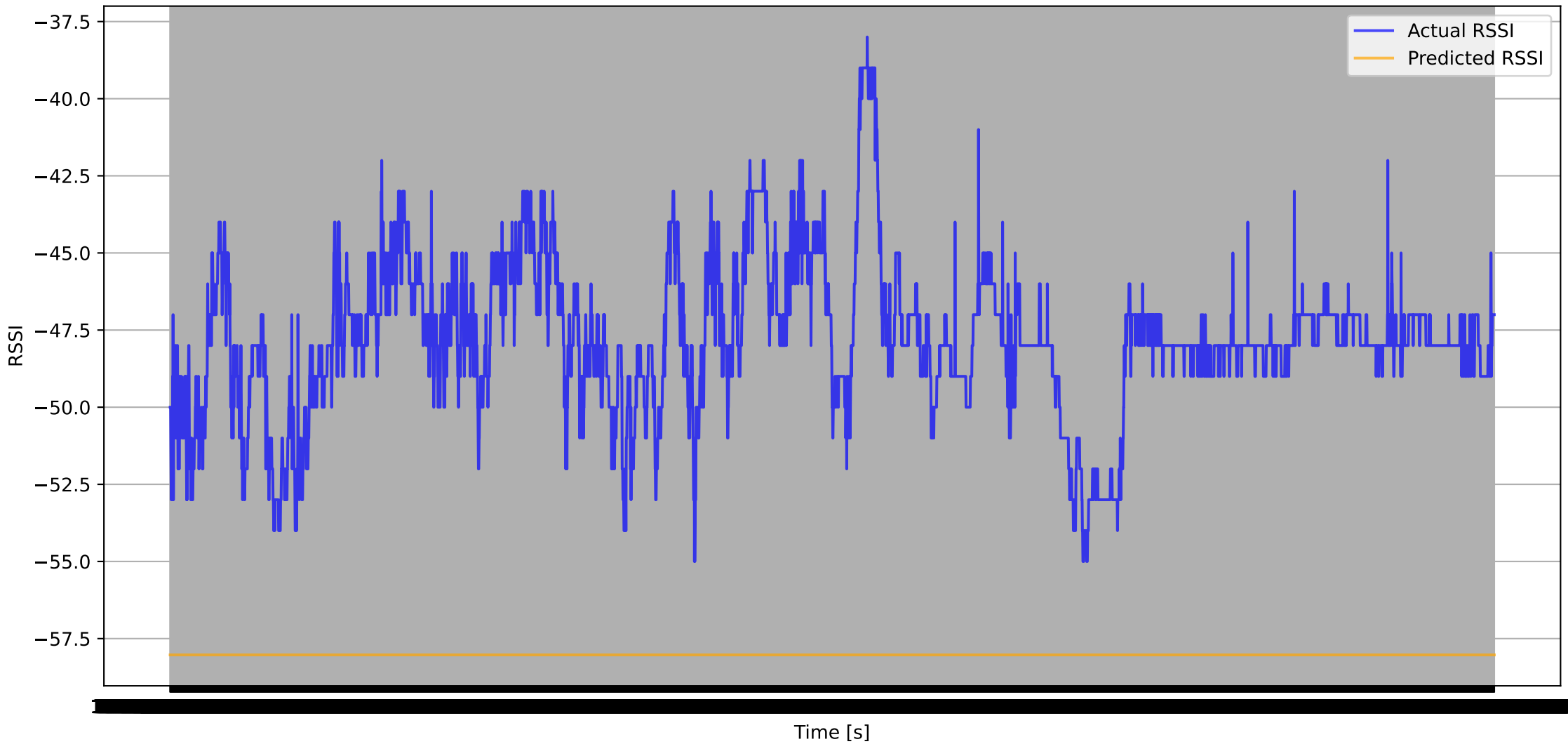
MAE: 0.3528

MSE: 0.1317

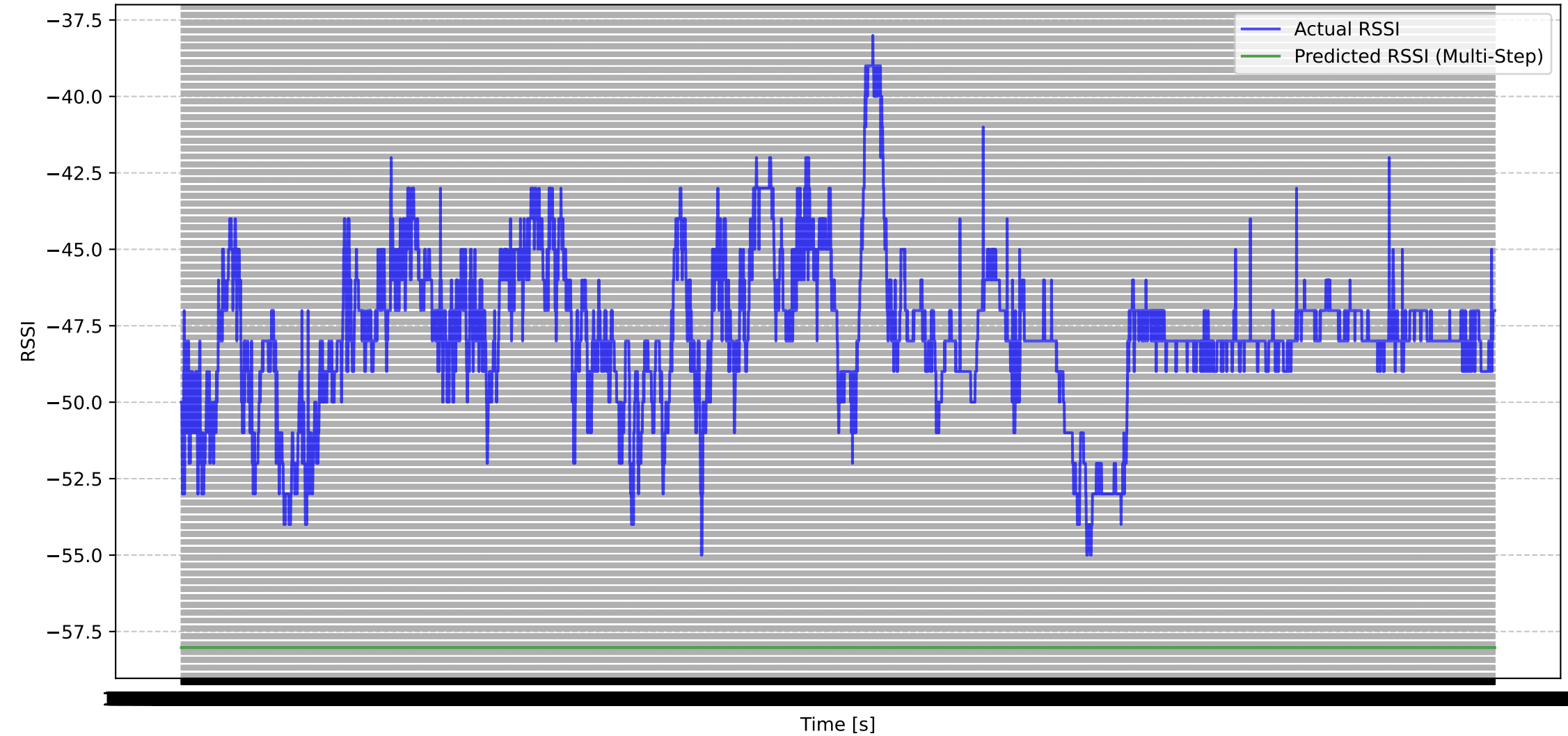
RMSE: 0.3629

R2 Score: -17.2333

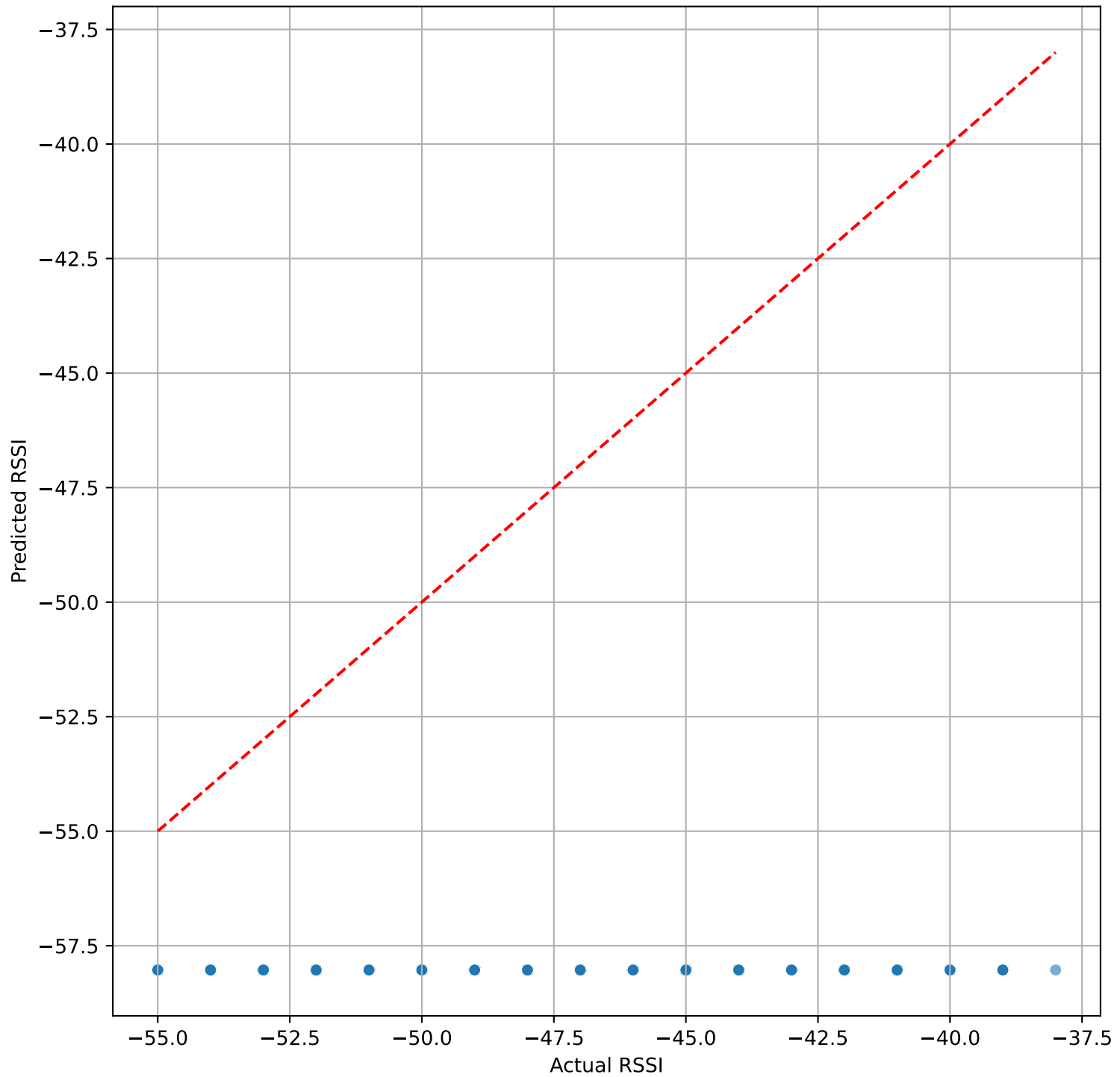
Actual vs Predicted RSSI (First Step)



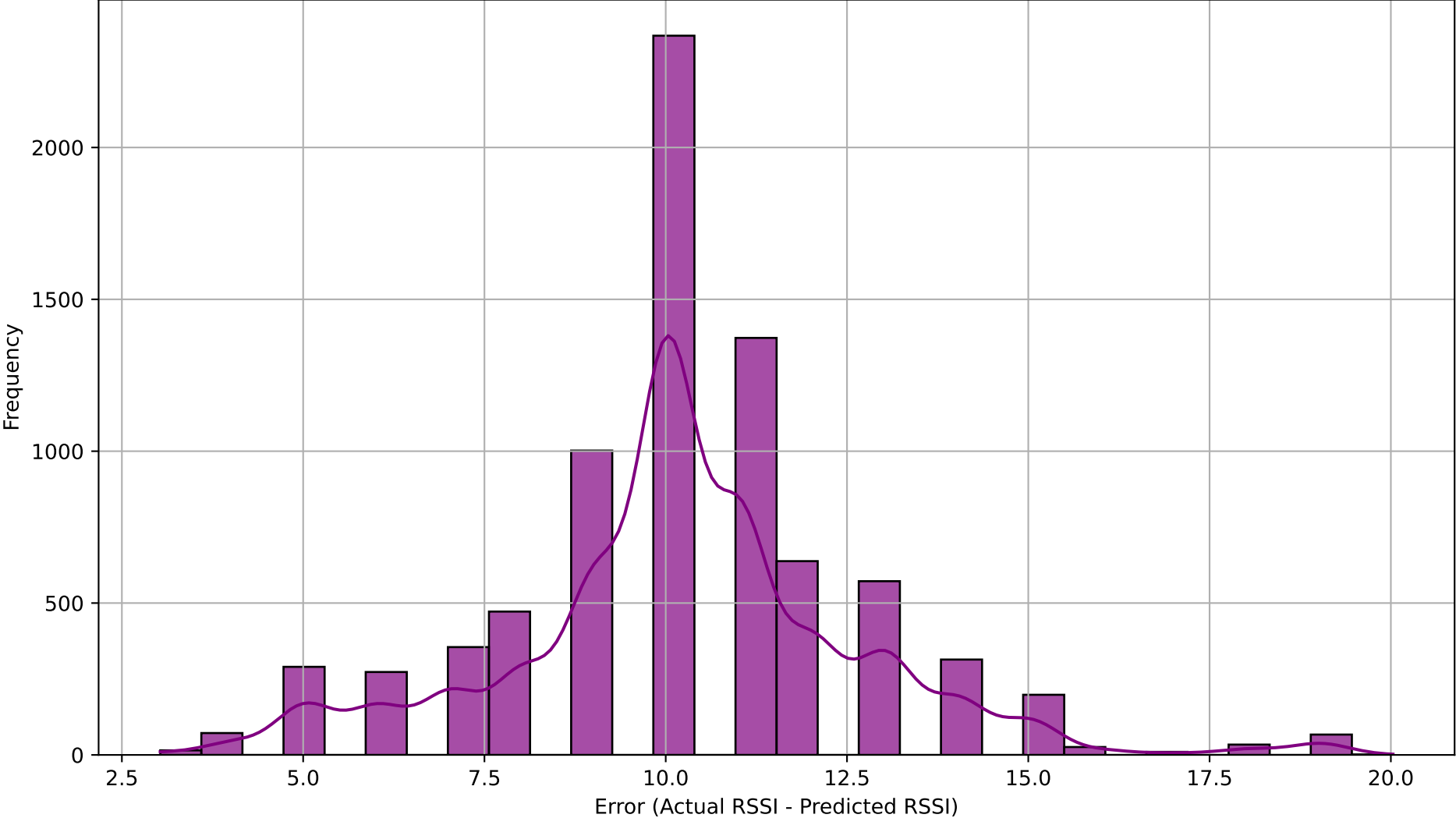
Actual vs Predicted RSSI (First Step vs Multi-Step)



Actual vs Predicted RSSI (First Step)

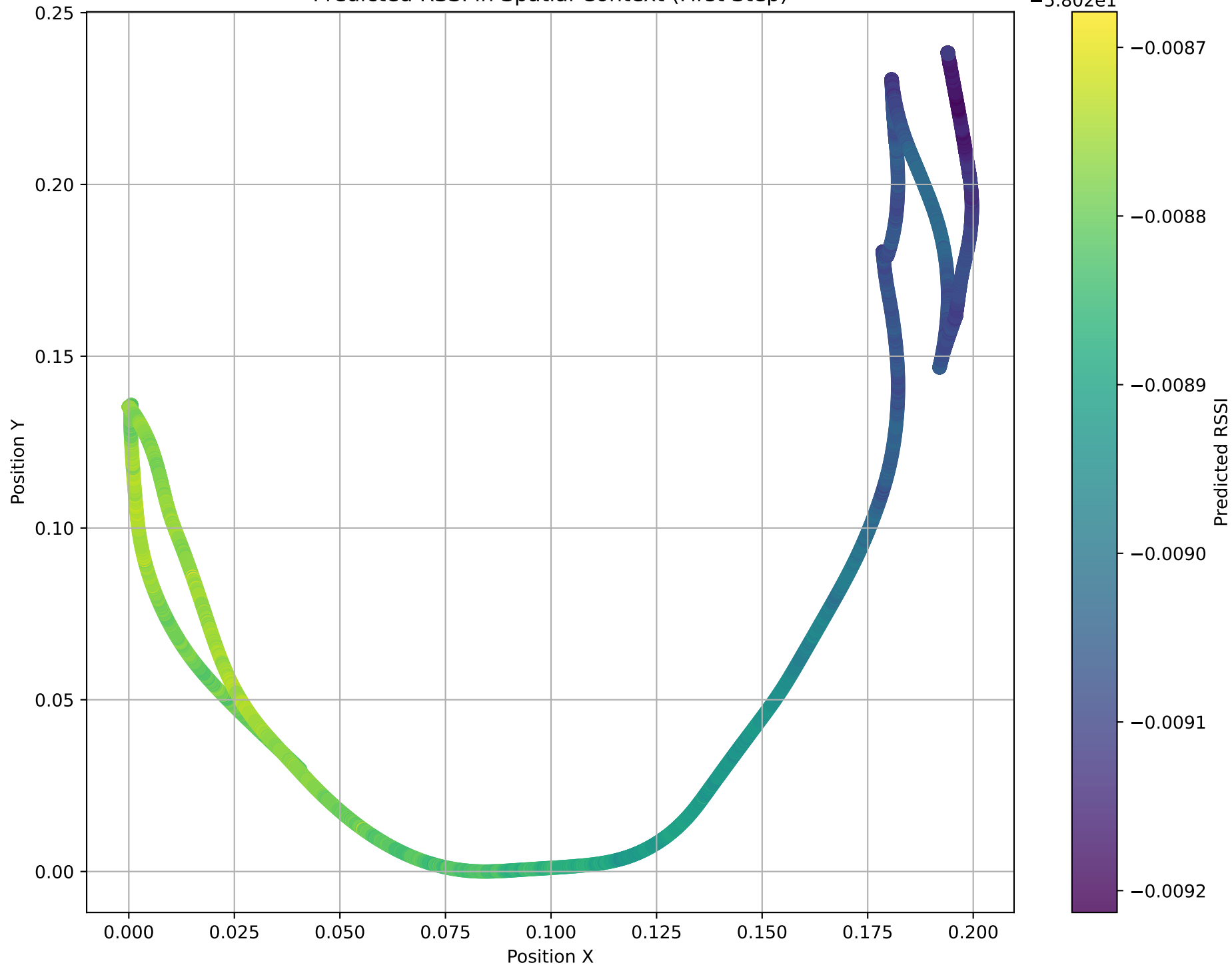


Distribution of Prediction Errors (First Step)



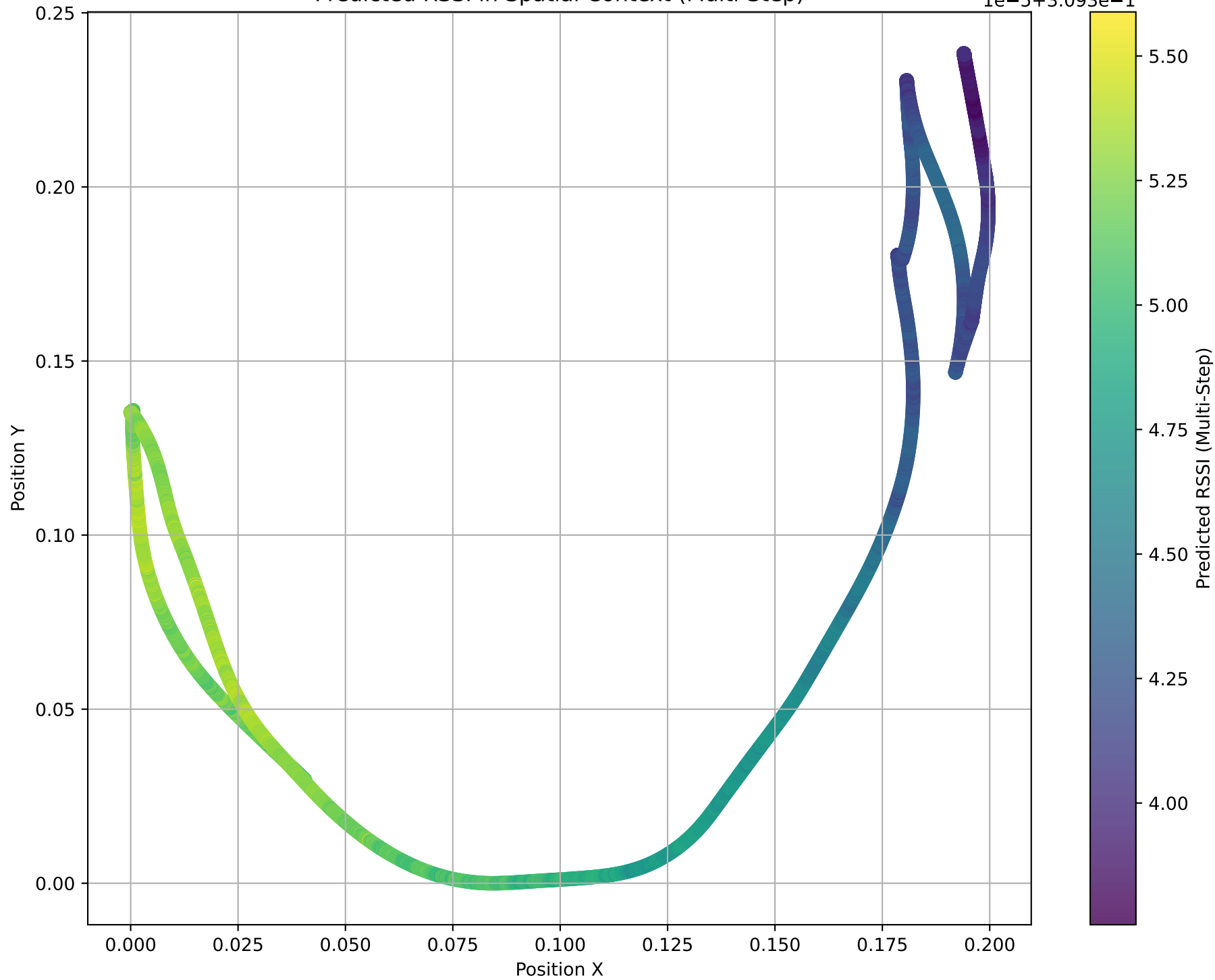
Predicted RSSI in Spatial Context (First Step)

$-5.802e1$

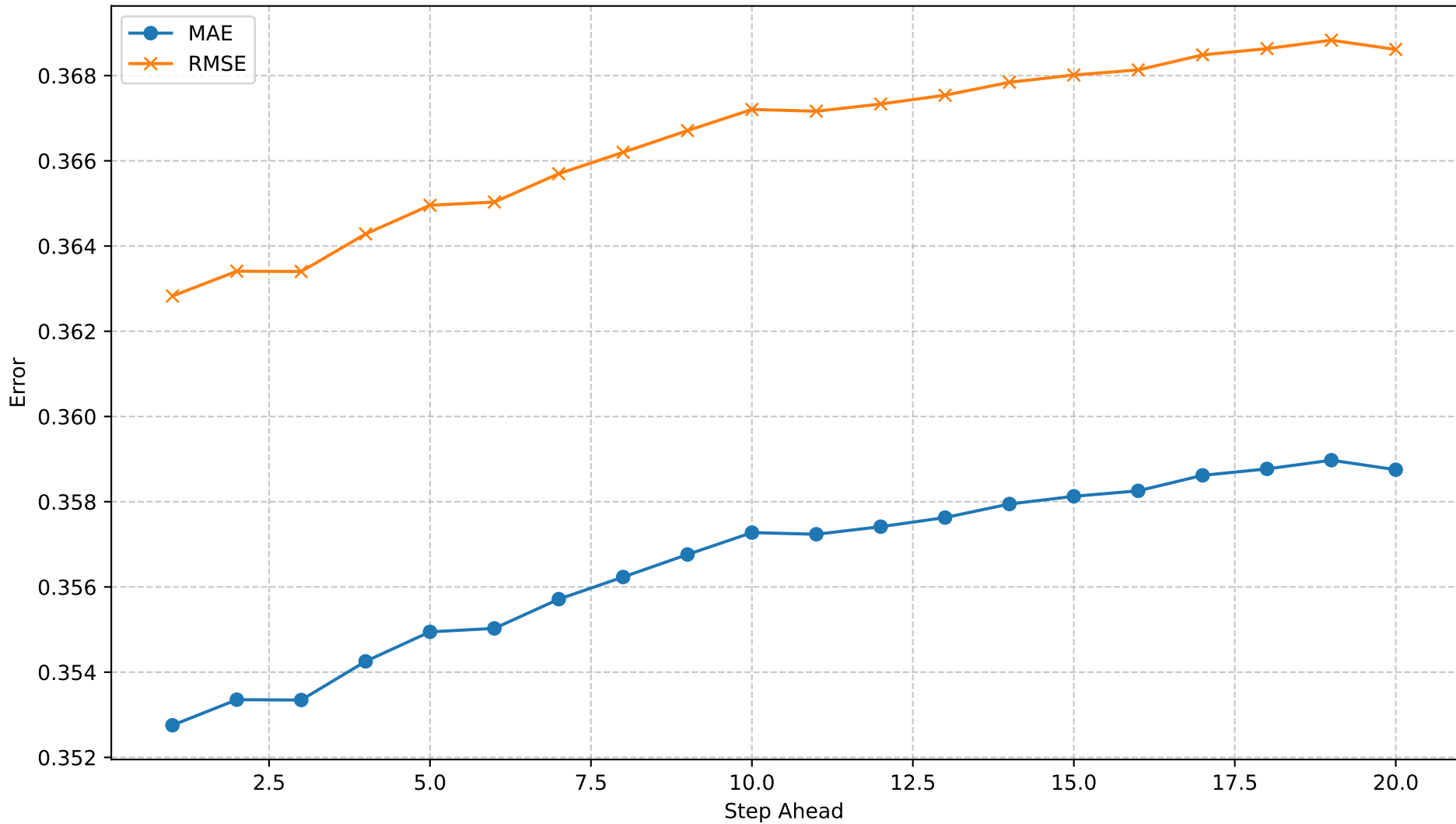


Predicted RSSI in Spatial Context (Multi-Step)

$1e-5+3.093e-1$



Multi-Step Forecast Error by Step



Resumo Automático:

- Forecast Steps configurados: 20
- Passos com MAE ≤ 2.0 : 20 passo(s)
- MAE no último passo: 0.3588

[V] O modelo mantém boa acurácia durante quase toda a sequência de previsões.